

THREAT INTELLIGENCE INTEGRATION WITH OPENCTI AND WAZUH

Building an Intelligence-Driven Detection Pipeline

Environment: Virtual Cybersecurity Lab

Platform: OpenCTI

SIEM: Wazuh

Threat Intelligence: AlienVault OTX and ThreatFox

Detection Use Case: SSH Password Guessing

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Technical Project Report

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Executive Summary

This project extended an existing Wazuh security monitoring laboratory by introducing OpenCTI as a dedicated threat intelligence platform. The objective was to demonstrate an operational workflow in which external threat intelligence could be collected, organized, distributed and applied to a real SIEM detection.

AlienVault OTX and ThreatFox were integrated as external intelligence sources. Their successful ingestion was verified through completed connector operations. A filtered OpenCTI collection named Wazuh-CTI-Indicators was then exposed through TAXII 2.1. A Python-based consumer retrieved the indicators, parsed IPv4, domain and SHA-256 values, and maintained a local cache for fast matching.

The final stage applied the intelligence to an existing SSH password-guessing detection in Wazuh. The baseline rule identified repeated failed SSH authentication attempts. The improved workflow preserved that detection and added OpenCTI context when the source IP matched an intelligence indicator.

Project area	Result
OpenCTI deployment	Operational
External feeds	AlienVault OTX and ThreatFox successfully ingested
TAXII	Validated collection and object retrieval
Local CTI cache	14,212 indicators: 5,173 SHA-256, 2,658 IPv4, 6,381 domains
Wazuh enrichment	Operational
Applied use case	SSH Password Guessing – T1110.001
Final CTI-aware rule	100105, level 12

Introduction

Security operations teams increasingly depend on both security telemetry and external threat intelligence when investigating suspicious activity. A SIEM can collect and correlate events from endpoints and other sources, but the resulting alert may not contain enough information to establish the wider significance of an indicator.

Threat intelligence can add context about infrastructure, malware, domains, hashes and other observable objects. OpenCTI provides a platform for

managing this intelligence using a structured data model, while Wazuh provides security monitoring, rule-based detection and alerting.

This project connects the two functions. Rather than treating threat intelligence as a separate repository, I designed a workflow in which OpenCTI intelligence could be consumed by a Wazuh integration and used to enrich an existing detection.

The implementation deliberately remained narrow. The objective was not to demonstrate every OpenCTI feature, but to build a working intelligence-to-detection path that could be tested and explained.

The final use case focused on SSH password guessing. This was selected because the detection already existed in the earlier detection engineering project and provided a clear opportunity to demonstrate how CTI can add investigative context.

Problem Statement

A Wazuh alert identifying repeated failed SSH authentication attempts can show the source IP, target endpoint, account and timing of the activity. However, without additional intelligence, the alert does not automatically show whether the source infrastructure is represented in a CTI collection.

This creates an analyst-context problem. The analyst may need to perform a separate intelligence lookup before deciding how much investigative attention the alert deserves.

The aim of this project was to build and validate an OpenCTI-based threat intelligence workflow integrated with Wazuh and demonstrate its practical value through an improved detection use case.

Project Scope and Objectives

1. Deploy and configure a functional OpenCTI instance.
2. Integrate multiple external threat intelligence feeds.
3. Validate successful ingestion of threat intelligence into OpenCTI.
4. Create a filtered OpenCTI collection containing relevant indicators.
5. Expose the collection through TAXII 2.1.
6. Develop a Python-based CTI consumer and local indicator cache.
7. Integrate the CTI enrichment process with Wazuh.
8. Enrich a Wazuh detection with OpenCTI context.
9. Improve an existing SSH password-guessing detection workflow.
10. Validate the improved workflow using controlled testing.

11. Document the implementation, troubleshooting process and lessons learned.

- OpenCTI threat intelligence management
- AlienVault OTX and ThreatFox ingestion
- TAXII 2.1 indicator distribution
- IPv4, domain and SHA-256 indicator matching
- Python-based enrichment and local caching
- Wazuh rule-based detection and alert enrichment
- SSH authentication activity
- MITRE ATT&CK mapping for T1110.001

Architecture and Environment

The final architecture places OpenCTI between external threat intelligence sources and the existing Wazuh monitoring environment. OpenCTI provides the intelligence management and filtering layer, while the Python consumer provides the controlled interface between CTI data and Wazuh.

The implementation separates the OpenCTI deployment from the Wazuh integration code. The OpenCTI Docker deployment was maintained under `/home/kali/opencti/docker`, while the CTI consumer and enrichment project was maintained under `/opt/opencti-wazuh`.

Component	Role	Relevant location / identifier
OpenCTI	Threat intelligence platform	Docker deployment
AlienVault OTX	External CTI source	OpenCTI connector
ThreatFox	External CTI source	OpenCTI connector
TAXII 2.1	CTI distribution	OpenCTI collection
Wazuh Manager	SIEM and detection engine	Wazuh Manager
Ubuntu Endpoint	SSH detection target	192.168.56.109
Kali Linux	Controlled testing source	Lab testing host

Architecture Diagram

The following diagram represents the implemented intelligence flow. External feeds populate OpenCTI, selected indicators are exposed through TAXII, the Python consumer maintains a local cache, and Wazuh uses that cache to enrich matching detections.

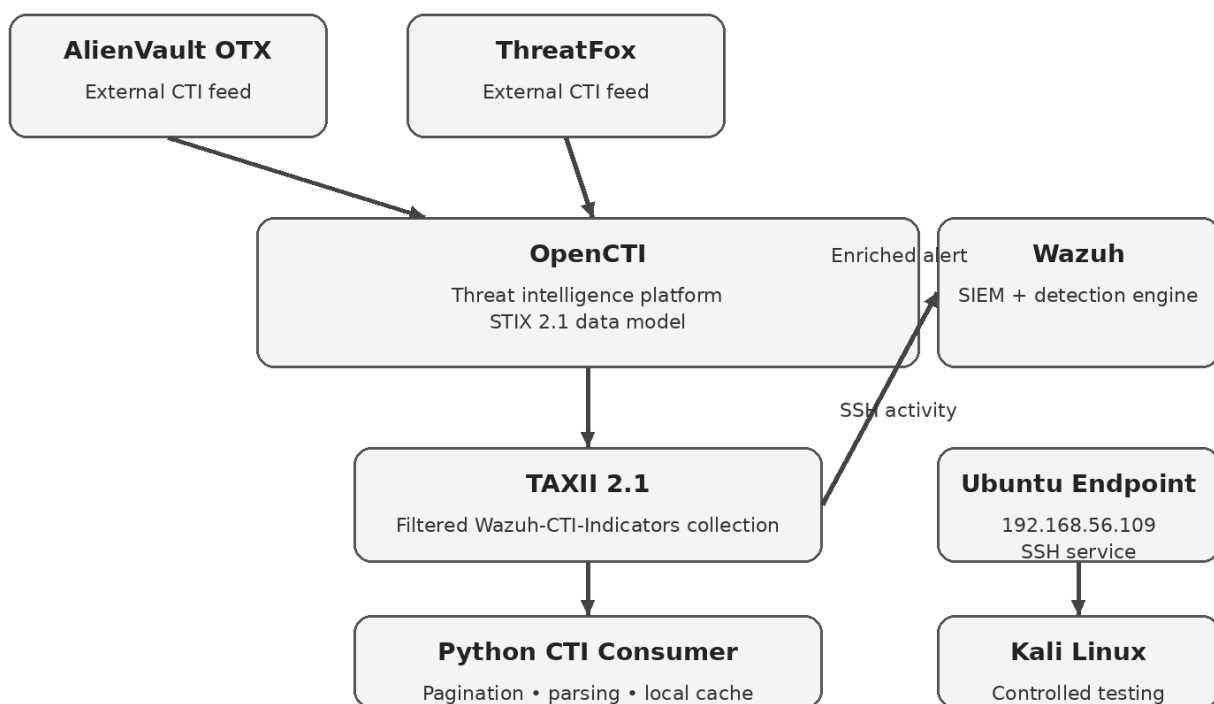


Figure: OpenCTI–TAXII–Python–Wazuh architecture used in the project.

The architecture deliberately keeps intelligence collection separate from detection logic. This makes the enrichment layer reusable for multiple Wazuh rules and indicator types.

Lab Components

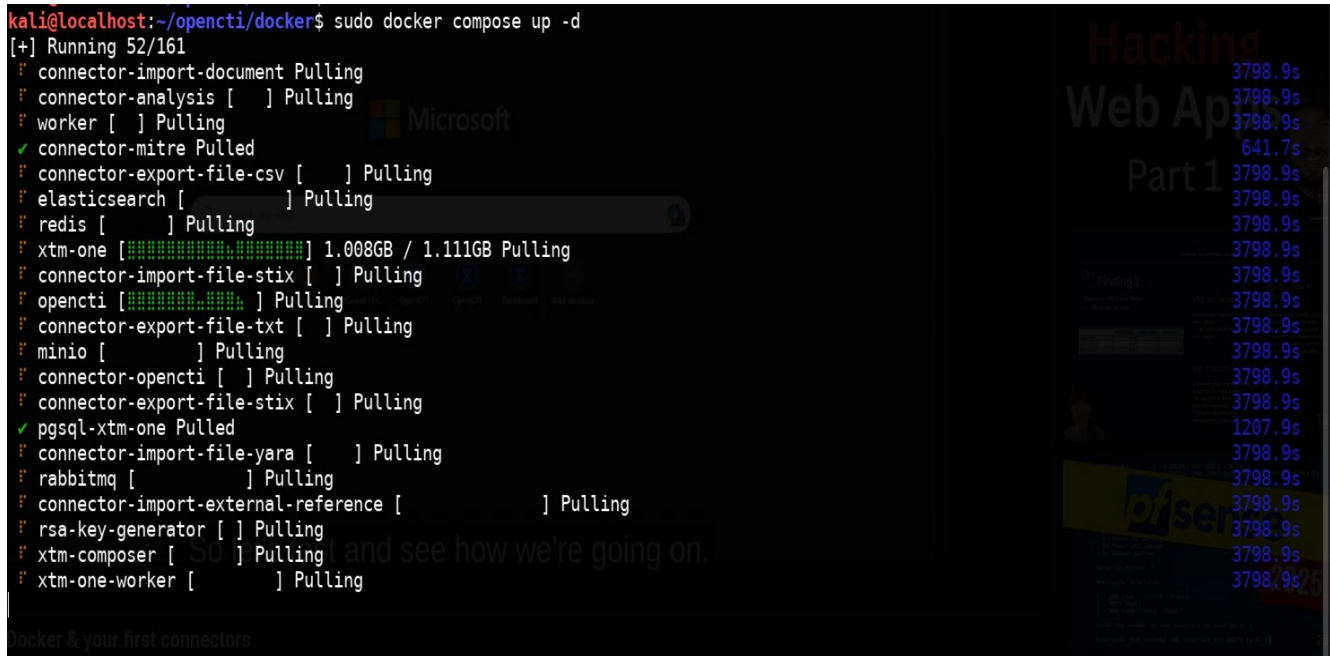
The project used the following main systems and technologies:

Technology	Use in project
OpenCTI	Central CTI management platform
Wazuh	SIEM, log collection and detection engine
Docker	OpenCTI deployment
TAXII 2.1	Threat intelligence distribution
STIX 2.1	Threat intelligence representation
AlienVault OTX	Broad community intelligence feed
ThreatFox / Abuse.ch	Malware and infrastructure intelligence
MITRE ATT&CK	Detection technique mapping
Ubuntu	Endpoint used for SSH detection
Kali Linux	Controlled security testing host

OpenCTI Deployment and Platform Setup

OpenCTI Deployment

OpenCTI was deployed using Docker. The deployment was kept separate from the Wazuh integration project so that platform configuration and SIEM-side enrichment code could be managed independently.



```
kali@localhost:~/opencti/docker$ sudo docker compose up -d
[+] Running 52/161
 connector-import-document Pulling
 connector-analysis [ ] Pulling
 worker [ ] Pulling
 connector-mitre Pulled
 connector-export-file-csv [ ] Pulling
 elasticsearch [ ] Pulling
 redis [ ] Pulling
 xtm-one [#####] 1.008GB / 1.111GB Pulling
 connector-import-file-stix [ ] Pulling
 opencti [#####] Pulling
 connector-export-file-txt [ ] Pulling
 minio [ ] Pulling
 connector-opencti [ ] Pulling
 connector-export-file-stix [ ] Pulling
 pgsql-xtm-one Pulled
 connector-import-file-yara [ ] Pulling
 rabbitmq [ ] Pulling
 connector-import-external-reference [ ] Pulling
 rsa-key-generator [ ] Pulling
 xtm-composer [ ] Pulling
 xtm-one-worker [ ] Pulling
```

The deployment provided the platform used for connector management, indicator storage, collection filtering and TAXII access.

OpenCTI Platform Setup

After deployment, the OpenCTI platform was configured to receive external intelligence through connectors. The project did not attempt to configure every available OpenCTI connector or data model feature.

The configuration focused on producing a controlled collection of indicators suitable for downstream Wazuh enrichment.

- External connectors were enabled and validated.
- A dedicated collection named Wazuh-CTI-Indicators was created.
- Indicator filtering was limited to relevant observable types.
- OpenCTI scores were copied to confidence for downstream use.
- Collection access was restricted to administrators.
- Inferences were not included in the collection.



Threat Intelligence Feed Integration

Feed Selection

I selected two primary feeds for the final implementation: AlienVault OTX and ThreatFox. Both provide indicator-oriented intelligence, but they originate from different ecosystems and provide useful complementary coverage.

Feed	Reason for selection	Role in project
AlienVault OTX	Broad community-driven CTI coverage	Primary external intelligence source
ThreatFox	Malware and infrastructure intelligence through Abuse.ch	Secondary intelligence source

The use of two feeds also allowed the project to demonstrate that OpenCTI was functioning as a collection and management layer rather than being tied to a single intelligence provider.

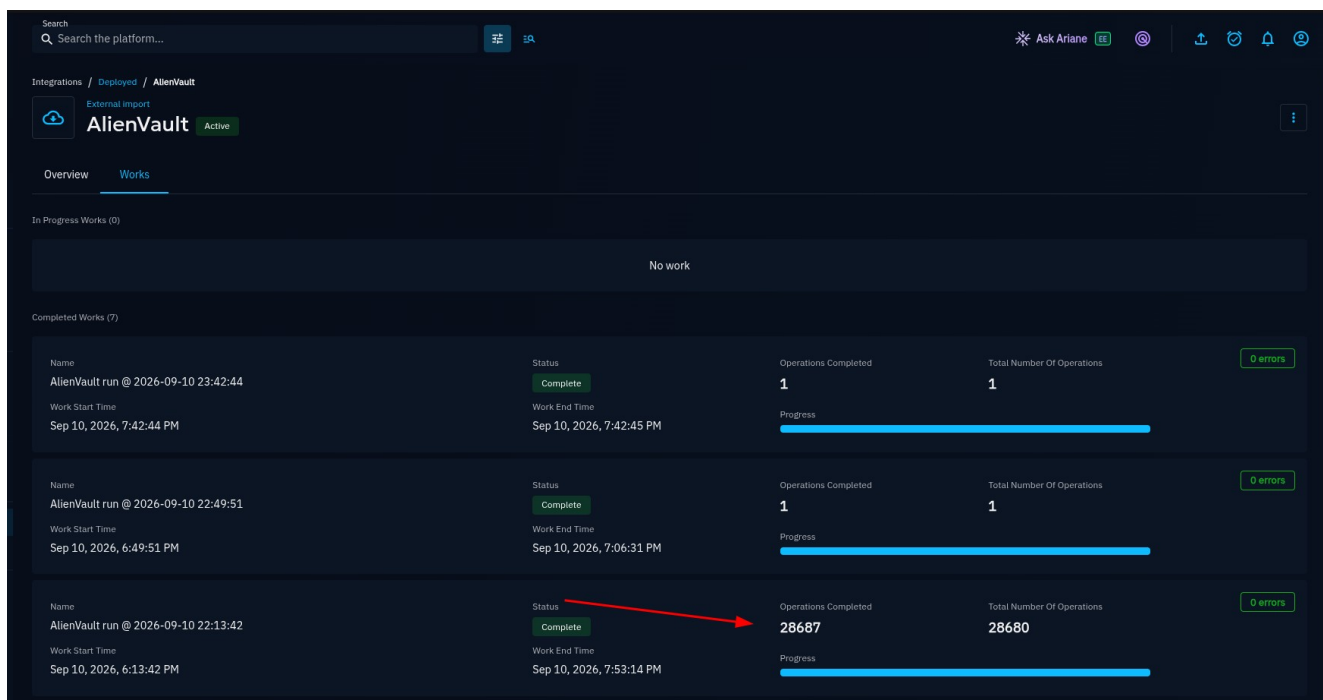
AlienVault OTX Integration

The AlienVault OTX connector was configured in OpenCTI and executed successfully. The completed operation provided evidence that the feed was not merely configured but actually processed by the platform.

Metric	Observed result
Operations Completed	28,687
Total Operations	28,680
Status	Complete
Errors	0

```

kali@localhost:~/opencti/docker$ sudo docker compose up -d
[+] Running 9/9
  ✓ connector-alienvault Pulled
    ✓ 55afalecc21d Already exists
    ✓ 47393b668ce5 Already exists
    ✓ 2014fd3d47b2 Already exists
    ✓ ed62c91ccd94 Already exists
    ✓ 0341ff29c0b8 Pull complete
    ✓ 6146c474985c Pull complete
    ✓ 6a717727a356 Pull complete
    ✓ 3a2e8b699ad6 Pull complete
[+] Running 24/24
  ✓ Container xtm-redis-1 Healthy
  ✓ Container xtm-elasticsearch-1 Healthy
  ✓ Container xtm-minio-1 Healthy
  ✓ Container xtm-pgsql-xtm-one-1 Healthy
  ✓ Container xtm-rabbitmq-1 Healthy
  ✓ Container xtm-rsa-key-generator-1 Healthy
  ✓ Container xtm-xtm-one-1 Healthy
  ✓ Container xtm-xtm-one-worker-1 Running
  ✓ Container xtm-opencti-1 Healthy
  ✓ Container xtm-connector-import-file-yara-1 Running
  ✓ Container xtm-connector-export-file-csv-1 Running
  ✓ Container xtm-connector-export-file-stix-1 Running
  ✓ Container xtm-connector-mitre-1 Running
  ✓ Container xtm-connector-import-external-reference-1 Running
  ✓ Container xtm-xtm-composer-1 Running
  ✓ Container xtm-connector-alienvault-1 Started
  ✓ Container xtm-connector-export-file-txt-1 Running
  ✓ Container xtm-connector-import-document-1 Running
  ✓ Container xtm-connector-analysis-1 Running
  ✓ Container xtm-connector-opencti-1 Running
  ✓ Container xtm-connector-import-file-stix-1 Running
  ✓ Container xtm-worker-1 Running
  ✓ Container xtm-worker-2 Running
  ✓ Container xtm-worker-3 Running
  
```

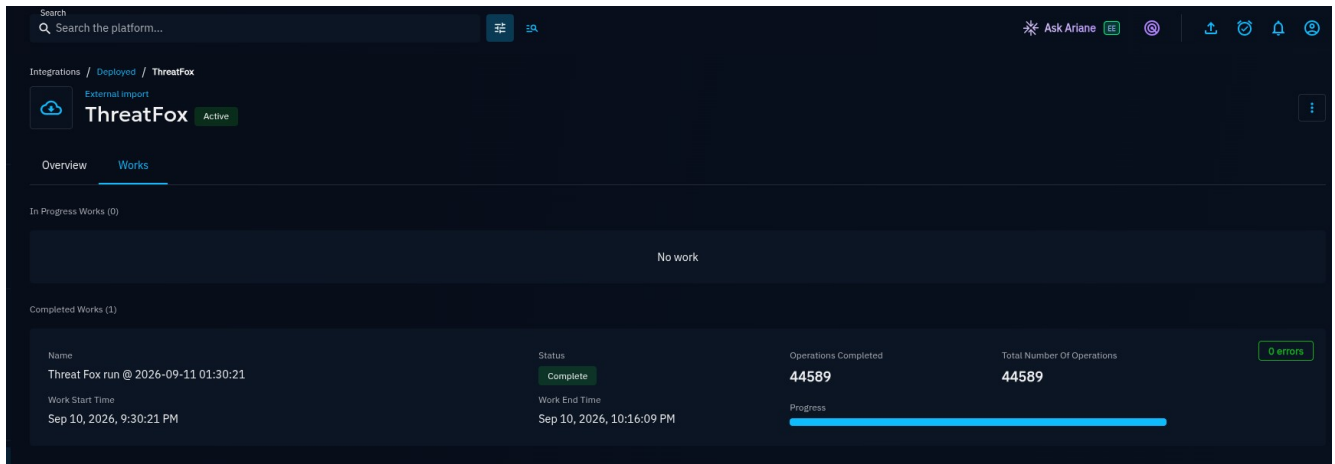


ThreatFox Integration

ThreatFox was configured as the second primary feed. It was selected because it provides malware-related indicators and infrastructure intelligence through the Abuse.ch ecosystem.

Metric	Observed result
Operations Completed	44,589
Total Operations	44,589
Status	Complete
Errors	0

```
kali@localhost:~/opencti/docker$ sudo docker compose up -d
[sudo] password for kali:
[+] Running 8/8
  connector-threatfox Pulled
  55afalecc21d Already exists
  47393b668ce5 Already exists
  2014fd3d47b2 Already exists
  ed62c91ccd94 Already exists
  ee23586920e3 Pull complete
  4f4fb700ef54 Pull complete
  21d48c33f2b3 Pull complete
[+] Running 26/26
connector-threatfox:
  image: opencti/connector-threatfox:latest
  environment:
    - OPENCTI_URL=http://localhost
    - OPENCTI_TOKEN=ChangeMe
    - CONNECTOR_ID=ChangeMe
    - CONNECTOR_NAME=ThreatFox
    - CONNECTOR_SCOPE=threatfox
    - CONNECTOR_LOG_LEVEL=error
```

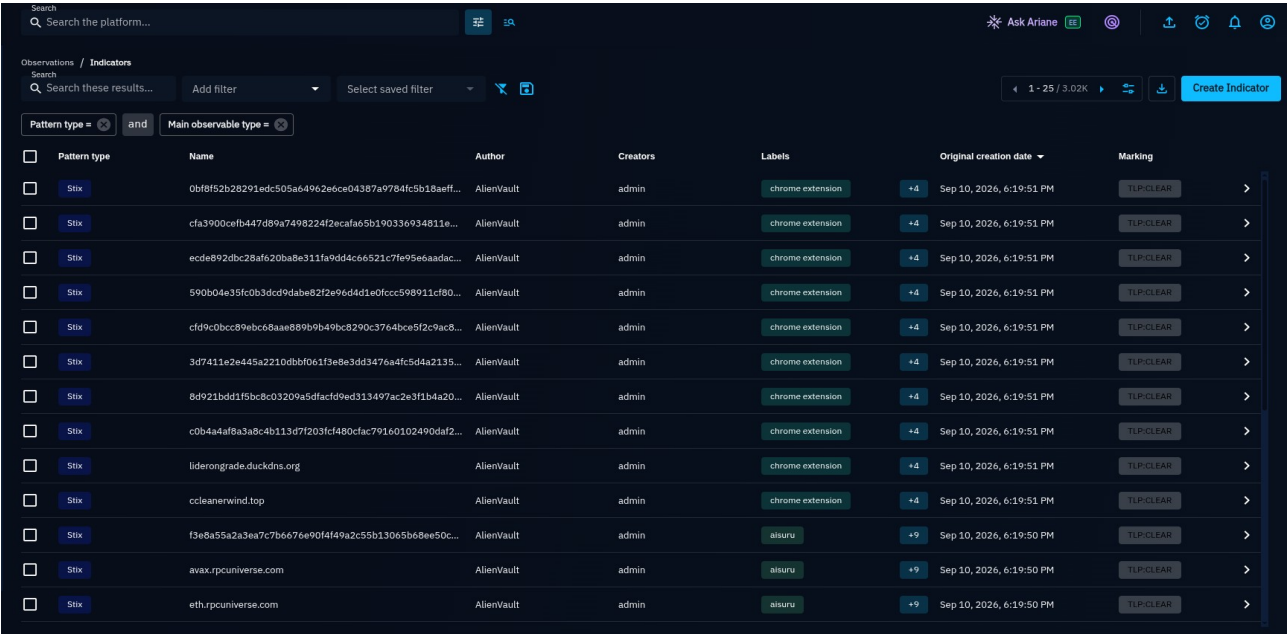


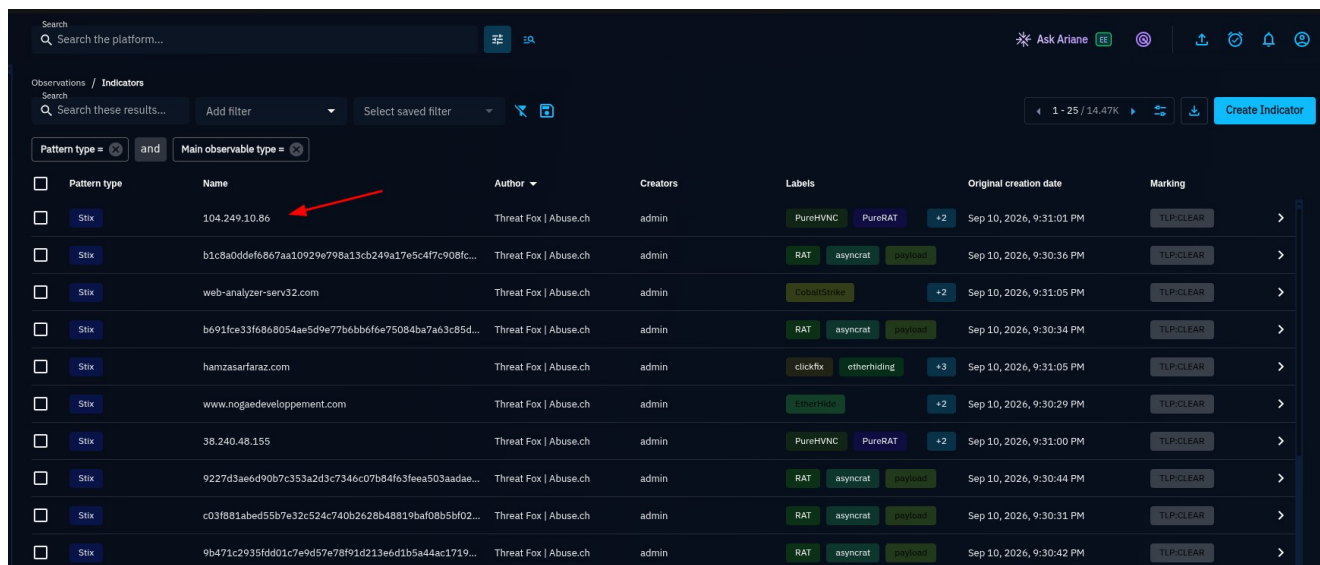
Feed Ingestion Results

Both selected feeds successfully populated the OpenCTI platform. The evidence is based on completed connector operations rather than simply showing that the connectors were enabled.

Feed	Status	Evidence to include
AlienVault OTX	Successful	Connector configuration + completed operation
ThreatFox	Successful	Connector configuration + completed operation

These results satisfied the project requirement to demonstrate live feed ingestion while keeping the final architecture focused on the two feeds that worked reliably in the lab.





CTI Data Selection and Mapping

Indicator Types

The enrichment pipeline focused on three indicator classes: IPv4 addresses, domains and SHA-256 file hashes. These types were selected because they map naturally to common Wazuh event fields and can be extracted reliably from STIX indicator patterns.

Indicator type	Example use	Wazuh relevance
IPv4	Network source or destination	Network and authentication events
Domain	DNS, URL or hostname context	Network and endpoint events
SHA-256	File or process artifact	Endpoint and malware-related events

The collection was filtered to indicators where the entity type was indicator and the main observable type was IPv4, file or domain.

OpenCTI Collection

A dedicated OpenCTI collection named Wazuh-CTI-Indicators was created for the downstream Wazuh workflow.

Collection setting	Configured value
Name	Wazuh-CTI-Indicators
Public	OFF
Accessible to	Administrators

Include inferences	OFF
Copy OpenCTI scores to confidence	ON
Collection ID	6ba526c6-9615-4663-b2a7-fc207d9dca7b
Filter	Indicator + IPv4 OR file OR domain

The screenshot shows the OpenCTI web interface. The left sidebar contains navigation links: Home, Dashboards, Investigations, PIR, Analyses, Cases, Events, Observations, Observables, Artifacts, Infrastructures, Threats, Arsenal, Techniques, Entities, Locations, Integrations, Data, and a Collapse button. The main content area is titled 'web-analyzer-serv32.com' and has tabs for Overview, Knowledge, Content, Analyses, Sightings, Data, and History. The 'Overview' tab is active, showing details for the indicator 'web-analyzer-serv32.com'. The indicator pattern is '[domain-name:value = 'web-analyzer-serv32.com']'. It has a score of 50/100 and a lifecycle status of 'No'. The valid from date is September 10, 2026 at 9:31:05 PM, and the valid until date is December 22, 2026 at 11:26:13 AM. The detection status is 'No'. The kill chain phases are empty. The indicator types are 'Domain', and the main observable type is 'Domain-Name'. The platforms are empty. The based on field is 'Domain Name'. The basic information section shows the marking as 'TLP:CLEAR', the author as 'Threat Fox | Abuse.Ch', the reliability as 'No', the confidence level as '2 - Probably True', the pattern type as 'Stix', the processing status as 'Disabled', the revoked status as 'No', and the labels as 'cobaltstrike', 'botnet_cc', and 'drb-ra'. The platform creation date is September 10, 2026 at 9:50:33 PM, and the creators are 'Admin'. The standard STIX ID is 'Indicator--890721bb-ca08-518c-8907-cb3bba25758f'. The original creation date is September 10, 2026 at 9:31:05 PM, and the modification date is September 10, 2026 at 9:50:33 PM.

The screenshot shows the Wazuh Discover interface. The left sidebar contains a search bar and a list of fields: Selected Fields (data.matched_field) and Available Fields (_index, agent.id, agent.name, data.agent.id, data.agent.name, data.arch, data.command, data.cti_confidence, data.cti_created, data.cti_indicator, data.cti_labels, data.cti_match, data.cti_modified, data.cti_pattern, data.cti_stix_id, data.cti_type, data.destination_ip, data.dpkg_status, data.dstuser, data.enrichment_timestamp, data.auid). The main content area is titled 'data.matched_field' and has tabs for DQL, Filter, and Last 48 hours. The 'Filter' tab is active, showing a bar chart of results for the query 'data.matched_field'. The chart shows a count of 5 hits for the time range 'Sep 23, 2026 @ 18:16:15.344 - Sep 25, 2026 @ 18:16:15.344 per Auto'. The hits are listed in a table below the chart:

Time	data.matched_field
Sep 25, 2026 @ 18:09:43.969	138a4c9cd617912c2269fae44bb12d57e926a36c2c62f25ee85a32cdf182212
Sep 25, 2026 @ 18:08:45.947	web-analyzer-serv32.com
Sep 25, 2026 @ 18:05:31.928	104.249.10.86
Sep 25, 2026 @ 14:07:46.940	104.249.10.86
Sep 25, 2026 @ 14:02:16.926	104.249.10.86

The expanded document is shown in a table below the hits:

Field	Value
_index	wazuh-alerts-4.x-2026.09.25
agent.id	000
agent.name	localhost
data.agent.id	000
data.agent.name	localhost
data.cti_confidence	50

STIX 2.1 and TAXII 2.1

OpenCTI represents threat intelligence using the STIX 2.1 model. TAXII 2.1 was used as the distribution interface between OpenCTI and the Python consumer.

```
localhost:8080/taxii2/root/collections/6ba526c6-9615-4663-b2a7-fc207d9dca7b/objects/

Pretty-print
spec_version: 2.1,
"type": "Indicator",
"extensions": {
  "extension-definition--ea279b3e-5c71-4632-ac08-831c66a786ba": {
    "extension_type": "property-extension",
    "id": "0980d912-889a-48d9-a588-0970c68edfe",
    "type": "Indicator",
    "created_at": "2026-09-10T22:34:03.173Z",
    "updated_at": "2026-09-10T22:34:03.214Z",
    "is_inferred": false,
    "creator_ids": [
      "88ec0c6a-13ce-5e39-b486-354fe4a7084f"
    ],
    "labels_ids": [
      "03787718-4c32-4172-bf8a-76872630f10f",
      "a3f8115c-083c-4de9-8588-45e357462d3",
      "9924299c-f43e-4a70-8a11-e3914154238e",
      "d599cf7d-d14b-4615-9997-8fb87650bf8f",
      "9236cc03-069f-4514-9a29-0136d2940325",
      "7e5cdeb4-1e08-4e9b-9222-bfda80c40af5",
      "ed0d708c-031f-403c-b9e1-ef934e530668"
    ],
    "created_by_ref_id": "03167f66-6b6f-4308-a2fd-565e63b06f2b",
    "created_by_ref_type": "Organization",
    "detection": false,
    "score": 85,
    "main_observable_type": "StixFile",
    "observable_values": [
      {
        "type": "StixFile",
        "hashes": {
          "SHA-256": "00759d29178baabcbe9682a953c64e179fd24d86dac0d6abdc8e5070216923f2"
        }
      }
    ],
    "extension-definition--322b8f77-262a-4cb8-a915-1e441e00329b": {
      "extension_type": "property-extension"
    }
  },
  "created": "2026-09-10T22:18:55.160Z",
  "modified": "2026-09-10T22:34:03.214Z",
  "revoked": false,
  "confidence": 85,
  "lang": "en",
  "labels": [
    "glutton",
    "glutton webshells",
    "secflow framework",
    "secbox",
    "ai-assisted intrusions",
    "fengtai government",
    "steganographic payloads"
  ],
  "object_marking_refs": [
    "marking-definition--613f2e26-407d-48c7-9eca-b8e91df99dc9"
  ],
  "created_by_ref": "identity--e52b2fa3-2af0-5e53-ad38-17d54b3d61cb",
  "name": "00759d29178baabcbe9682a953c64e179fd24d86dac0d6abdc8e5070216923f2",
  "pattern": "[file:hashes, 'SHA-256' = '00759d29178baabcbe9682a953c64e179fd24d86dac0d6abdc8e5070216923f2']",
  "pattern_type": "stix",
}
```

Search

Search the platform...

🔍

🔍

Data / Data sharing / TAXII collections

Search

Search these results...

Name	Description	Collection
Wazuh-CTI-Indicators	SIEM-to-OpenCTI Enrichment Pipeline: Connecting OpenCTI's Intelligence to Wazuh. This Stream is dedicated to feed my SIEM system with q	6ba526c6-9615-4663-b2a7-fc207d9dca7b

Update a TAXII collection

Name

Wazuh-CTI-Indicators

Description

SIEM-to-OpenCTI Enrichment Pipeline: Connecting OpenCTI's Intelligence to Wazuh. This Stream is dedicated to feed my SIEM system with q

Make this TAXII collection public and available to anyone

☐ Public collection

Accessible for

Administrators

Leave the field empty to grant all authenticated users

☐ Include inferences

☒ Copy OpenCTI scores to confidence level for indicators

Add filter

Entity type = indicator and Main observable type = ipv4 address or domain name or or and

Author = threat fox or alismavault

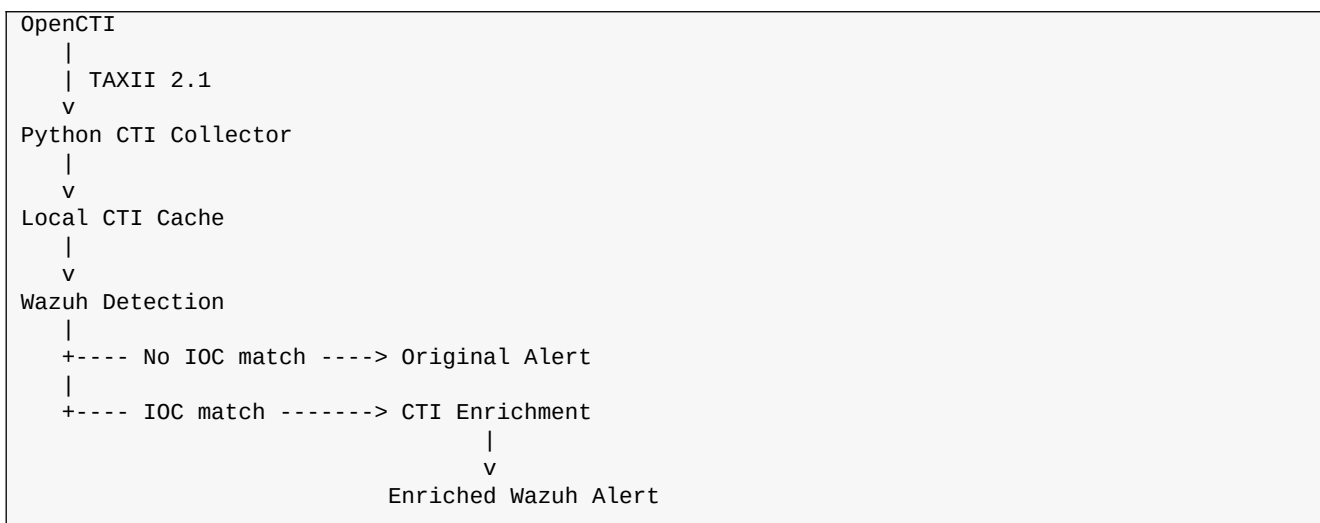
15

The TAXII endpoint returned HTTP 200 and exposed STIX indicator objects with pagination. This confirmed that the downstream consumer could retrieve intelligence programmatically.

Wazuh–OpenCTI Enrichment Pipeline

Pipeline Design

The enrichment pipeline was designed so that Wazuh remained responsible for security-event detection while OpenCTI supplied contextual intelligence. The Python component acted as the integration layer.



This architecture also means that the complete CTI cache is not itself a set of Wazuh alerts. An indicator becomes visible in Wazuh when a matching security event is processed and enriched.

CTI Collector

The Python collector was developed at `/opt/opencti-wazuh/cti_enrichment.py`. It used Python requests for HTTP communication and handled TAXII pagination so that the collection could be retrieved beyond a single response.

- Connect to the OpenCTI TAXII endpoint.
- Authenticate with a token stored outside the report and code.
- Retrieve STIX objects from the selected collection.
- Handle TAXII pagination.
- Extract IPv4, domain and SHA-256 patterns.
- Normalize the extracted indicator values.
- Store the resulting records in a local JSON cache.

Indicator Cache

The collector produced a local cache at `/opt/opencti-wazuh/cti_cache.json`. The cache was used to provide fast local lookups when Wazuh events were enriched.

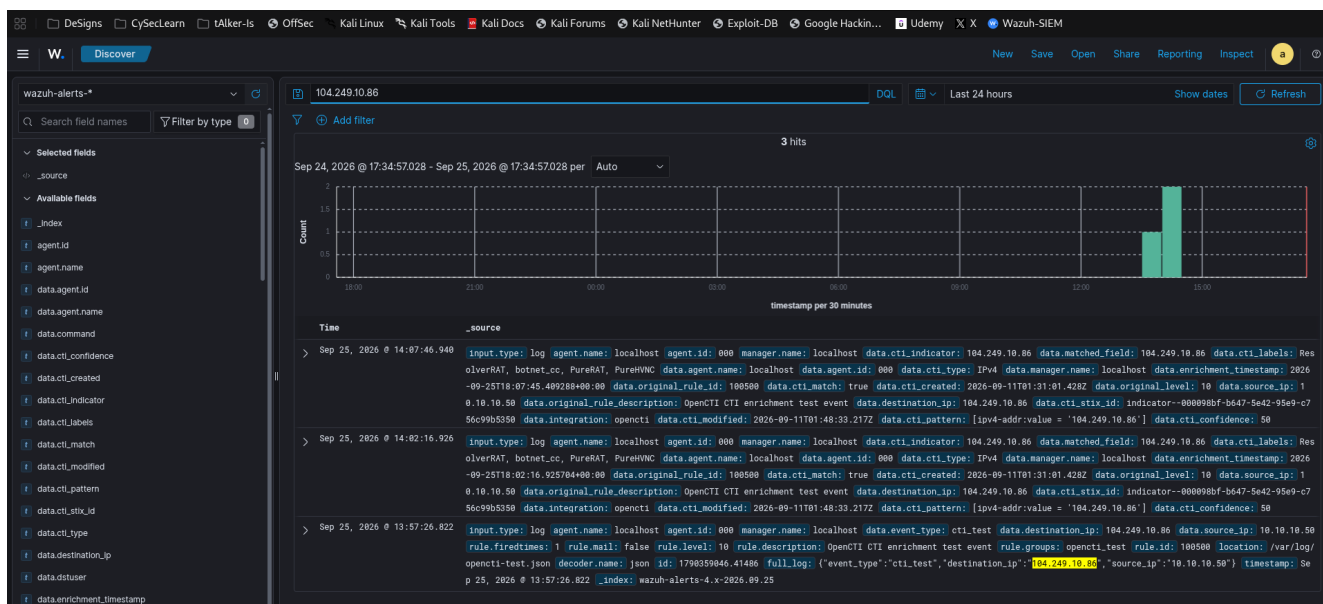
Indicator type	Count
SHA-256	5,173
IPv4	2,658
Domain	6,381
Total	14,212



Wazuh Integration

The first custom integration was implemented at `/var/ossec/integrations/custom-opencti`. It reads a Wazuh alert, identifies candidate observable fields, searches the local cache and writes an enriched JSON event.

Integration field	Purpose
integration	Identifies OpenCTI as the enrichment source
cti_match	Indicates that a cache match occurred
cti_indicator	Matched indicator
cti_type	Observable type
cti_confidence	OpenCTI confidence value
cti_labels	Associated CTI labels
cti_stix_id	STIX indicator identifier



Applied Detection Use Case

Detection Selection

For the applied use case, I selected the SSH password-guessing detection developed in the earlier detection engineering project. This provided a real existing detection that could be improved with CTI rather than creating an unrelated demonstration.

Rule ID: 100104

Level: 10

Frequency: 5

Timeframe: 120 seconds

MITRE ATT&CK: T1110.001

Condition: 5 failed SSH authentication attempts from the same source IP within 120 seconds.

The original rule was preserved as the baseline. A separate CTI-aware rule was introduced so that the before-and-after behavior could be demonstrated clearly.

MITRE ATT&CK Mapping

The selected detection maps to MITRE ATT&CK technique T1110.001 – Password Guessing, under the Credential Access tactic.

ATT&CK element	Mapping
Tactic	Credential Access
Technique	T1110 – Brute Force

Sub-technique	T1110.001 – Password Guessing
Detection behavior	Repeated failed SSH authentication attempts

MITRE ATT&CK describes T1110.001 as repetitive or iterative password guessing and includes SSH among the services commonly targeted by this behavior. MITRE also documents detection analytics involving repeated failed SSH authentication.

Home > Techniques > Enterprise > Brute Force > Password Guessing

Brute Force: Password Guessing

Other sub-techniques of Brute Force (4)

ID	Name
T1110.001	Password Guessing
T1110.002	Password Cracking
T1110.003	Password Spraying
T1110.004	Credential Stuffing

Adversaries with no prior knowledge of legitimate credentials within the system or environment may guess passwords to attempt access to accounts. Without knowledge of the password for an account, an adversary may opt to systematically guess the password using a repetitive or iterative mechanism. An adversary may guess login credentials without prior knowledge of system or environment passwords during an operation by using a list of common passwords. Password guessing may or may not take into account the target's policies on password complexity or use policies that may lock accounts out after a number of failed attempts.

Guessing passwords can be a risky option because it could cause numerous authentication failures and account lockouts, depending on the organization's login failure policies. ^[1]

Typically, management services over commonly used ports are used when guessing passwords. Commonly targeted services include the following:

- SSH (22/TCP)
- Telnet (23/TCP)
- FTP (21/TCP)

ID: T1110.001
Sub-technique of: [T1110](#)
① **Tactic:** [Credential Access](#)
① **Platforms:** Containers, ESXi, IaaS, Identity Provider, Linux, Network Devices, Office Suite, SaaS, Windows, macOS
Contributors: Microsoft Threat Intelligence Center (MSTIC); Mohamed Kmal
Version: 1.7
Created: 11 February 2020
Last Modified: 12 May 2026

[Version Permalink](#)

Password Guessing via Multi-Source Authentication Failure Correlation

Technique Detected: Password Guessing | T1110.001

ID: DET0551
Domains: Enterprise
Analytics: AN1521, AN1522, AN1523, AN1524, AN1525, AN1526
Version: 1.0
Created: 21 October 2025
Last Modified: 12 May 2026

[Version Permalink](#)

Analytics

Windows Linux macOS Identity Provider Network Devices SaaS

AN1521

Series of authentication failures (Event ID 4625) targeting the same or similar user accounts over time from one or more remote IPs

Log Sources

Data Component	Name	Channel
User Account Authentication (DC0002)	WinEventLog:Security	EventCode=4625

Mutable Elements

Field	Description
TimeWindow	Defines the period in which multiple failed attempts are aggregated (e.g., 10 minutes)

Baseline Detection – Before

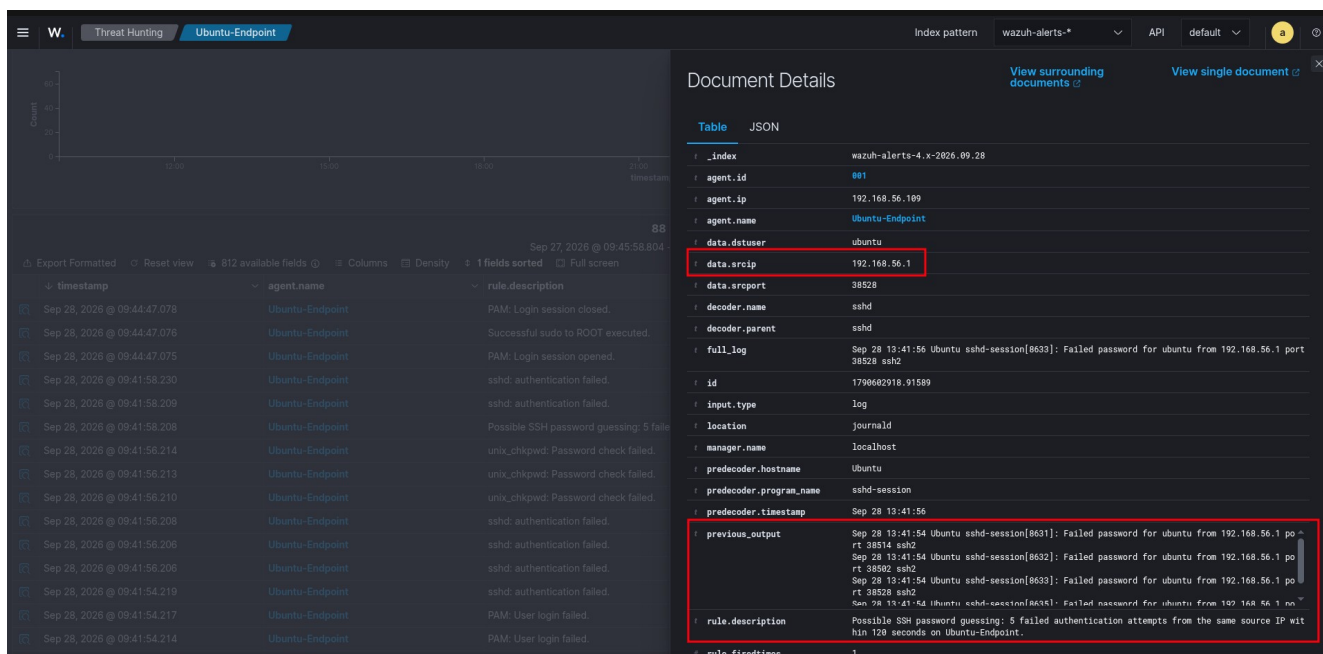
The baseline test used Hydra from Kali Linux against the Ubuntu endpoint at 192.168.56.109. The activity generated repeated failed SSH authentication attempts.

```
hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt -t 4 -Vv ssh://192.168.56.109
```

```
kali@localhost:~$ hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt -t 4 -Vv ssh://192.168.56.109
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations,
or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-28 09:41:50
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://192.168.56.109:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://ubuntu@192.168.56.109:22
[INFO] Successful, password authentication is supported by ssh://192.168.56.109:22
[ATTEMPT] target 192.168.56.109 - login "ubuntu" - pass "123456" - 1 of 14344399 [child 0] (0/0)
[ATTEMPT] target 192.168.56.109 - login "ubuntu" - pass "12345" - 2 of 14344399 [child 1] (0/0)
[ATTEMPT] target 192.168.56.109 - login "ubuntu" - pass "123456789" - 3 of 14344399 [child 2] (0/0)
[ATTEMPT] target 192.168.56.109 - login "ubuntu" - pass "password" - 4 of 14344399 [child 3] (0/0)
[ATTEMPT] target 192.168.56.109 - login "ubuntu" - pass "iloveyou" - 5 of 14344399 [child 2] (0/0)
[ATTEMPT] target 192.168.56.109 - login "ubuntu" - pass "princess" - 6 of 14344399 [child 1] (0/0)
[ATTEMPT] target 192.168.56.109 - login "ubuntu" - pass "1234567" - 7 of 14344399 [child 3] (0/0)
^C[ERROR] Received signal 2, going down ...
The session file ./hydra.restore was written. Type "hydra -R" to resume session.
kali@localhost:~$
```

Wazuh detected the activity using rule 100104. The resulting alert was level 10 and mapped to T1110.001. The baseline alert contained the security-event context but did not contain OpenCTI enrichment fields.



CTI-Aware Detection – After

The CTI-aware workflow preserved rule 100104 as the original detection and introduced rule 100105 as a second-stage alert for an OpenCTI match.

Rule ID: 100105

Level: 12

Description: CTI-enriched SSH password guessing: \$(cti_indicator)

MITRE ATT&CK: T1110.001

The processing sequence was: SSH authentication failures → rule 100104 → source-IP extraction → OpenCTI cache lookup → CTI match → enriched JSON event → rule 100105.

This design intentionally preserves the original detection. The CTI-aware rule adds context rather than replacing the initial detection.

CTI-Enriched Wazuh Alert

The following dashboard evidence shows the successful CTI-enriched SSH password-guessing alert. The event identifies the Ubuntu endpoint and displays the matched indicator, CTI confidence and labels.

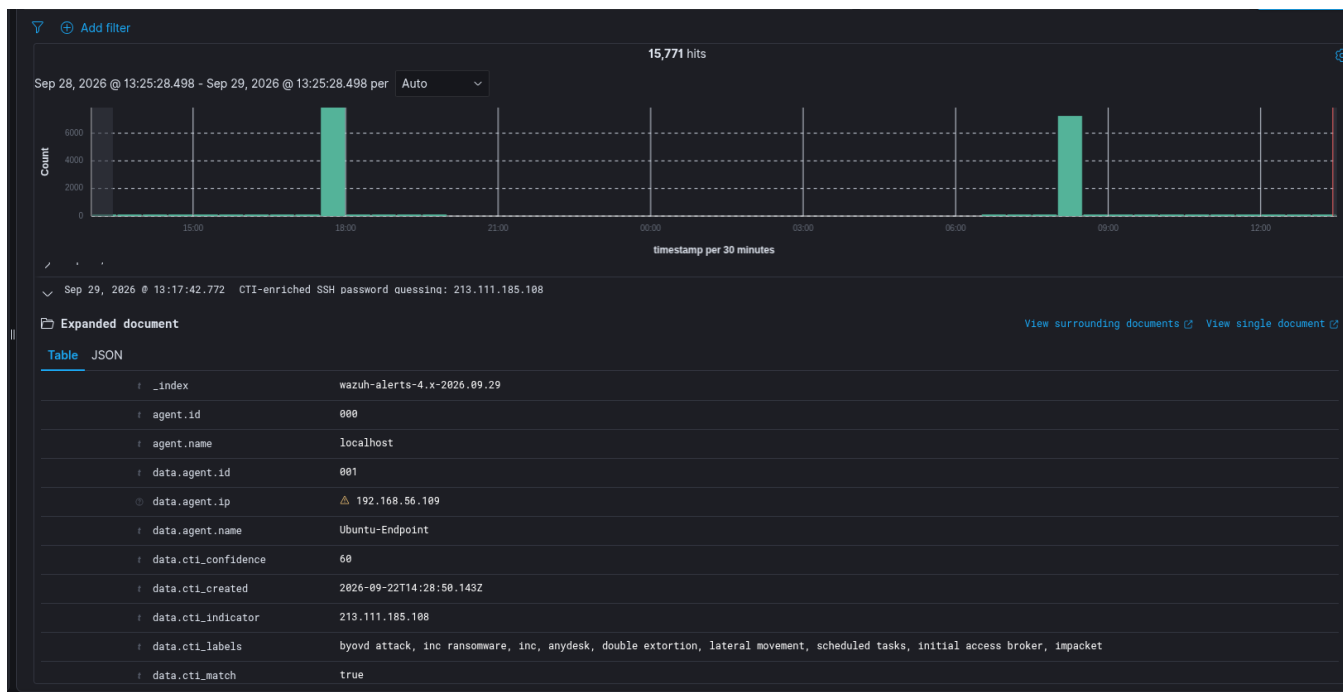


Figure: Wazuh dashboard showing the CTI-enriched SSH password-guessing alert.

Key evidence visible in the screenshot includes `cti_match=true`, `cti_indicator=213.111.185.108`, `cti_confidence=60` and the associated CTI labels.

Enrichment and Original Detection Details

The second dashboard view is particularly important because it shows the relationship between the CTI-enriched event and the original Wazuh detection.

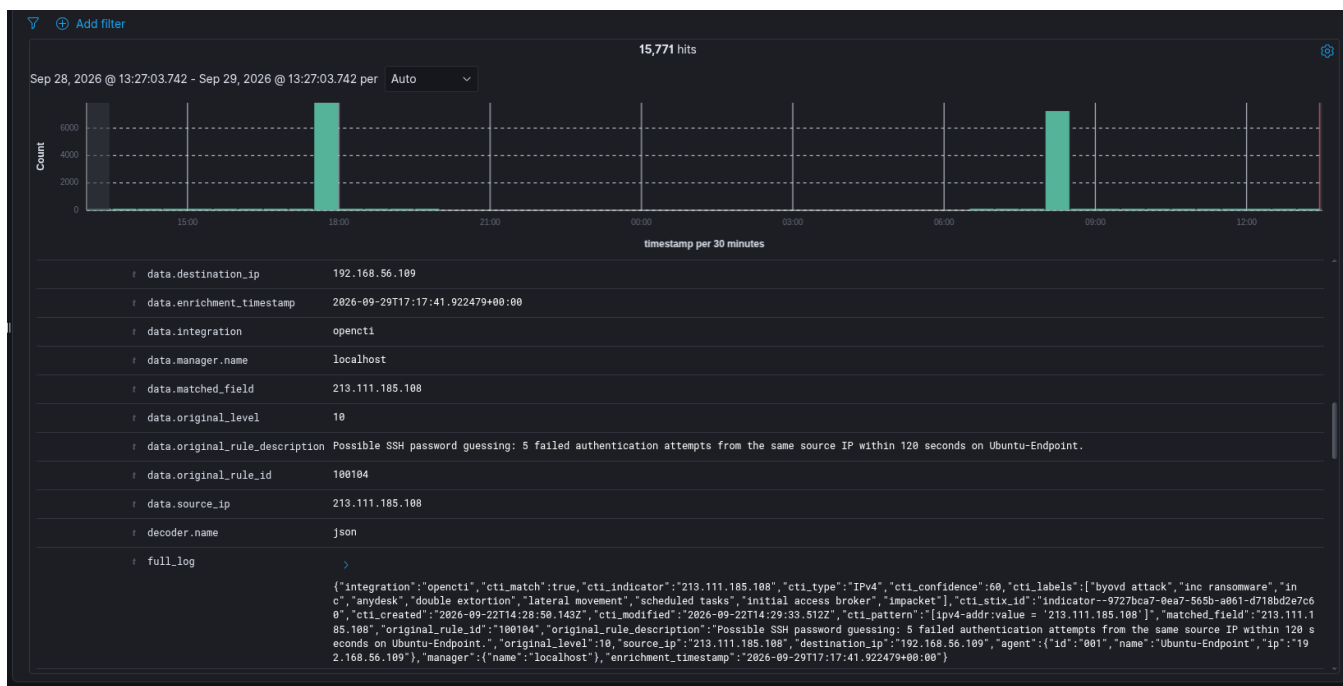


Figure: Wazuh dashboard showing OpenCTI enrichment fields and the preserved original detection.

The event retains original_rule_id=100104 and original_level=10 while adding the OpenCTI integration fields. This provides direct evidence that the CTI layer was applied to an existing detection.

Before and After Results

Attribute	Before	After
Detection	SSH password guessing	SSH password guessing + CTI match
Rule	100104	100105
Level	10	12
MITRE ATT&CK	T1110.001	T1110.001
OpenCTI context	None	Present
IOC	192.168.56.1	213.111.185.108
CTI confidence	—	60
CTI labels	—	Present
STIX ID	—	Present
Original rule preserved	Yes	Yes

Testing Interpretation

The testing demonstrated that the original detection remained functional and that a matching source IP could trigger a second-stage CTI-enriched alert.

The improvement demonstrated by the lab is contextual enrichment and alert prioritization. The project did not conduct a statistical false-positive study, so it would not be accurate to claim a measured percentage reduction in false positives.

The final alert gives an analyst additional information without requiring a separate manual intelligence lookup during the initial triage step.

Challenges and Troubleshooting

OpenCTI and Feed Configuration

The initial platform work required connector configuration, feed validation and testing. A VirusTotal Livehunt experiment also encountered a forbidden API response and was excluded from the final feed architecture.

Service Account and Output Permissions

The enrichment script initially attempted to modify output-file permissions after writing. This caused a permission problem when executed by the Wazuh integration service. The unnecessary chmod operation was removed and the script was kept compatible with the wazuh service account.

Manual Testing Method

The custom integration expected the alert JSON as a command-line argument. A pipe-to-stdin test therefore did not produce the expected output. Testing was corrected by invoking the integration with the alert as its first argument.

Lessons Learned

- Threat intelligence collection alone does not automatically improve detection. The intelligence must be connected to an operational workflow.
- Data structures matter. Understanding the actual STIX/TAXII and local cache representation was necessary to build reliable lookups.
- Indicator context is most useful when it appears alongside the security event being investigated.
- Confidence values should be reported precisely. A score of 60 should not automatically be described as high confidence.
- Separating the original detection from the CTI enrichment stage made troubleshooting and before/after validation much easier.
- Testing integrations under the same service account that will execute them is important for identifying permission problems early.
- The complete CTI cache is not the same thing as a set of Wazuh alerts. An indicator becomes operationally relevant when it matches an event.

Future Improvements

- Relationship-aware enrichment: retrieve relevant OpenCTI relationships so enriched alerts can include malware, threat actor, campaign or intrusion-set context where available.
- Real-time intelligence: investigate OpenCTI Live Streams or another event-driven distribution mechanism instead of relying only on periodic TAXII collection updates.
- Indicator lifecycle management: add cache refresh, expiration and stale-indicator handling.
- Source reliability: retain feed/source metadata and incorporate source reliability into prioritization.
- Asset context: combine CTI confidence with endpoint criticality and business context.
- Broader Wazuh coverage: reuse the same enrichment layer for DNS, web, endpoint process and file events.
- Detection analytics: test additional CTI-aware rules and measure false-positive behavior over a larger controlled dataset.

With additional time, I would also extend the enrichment model beyond indicator-level metadata. The current project deliberately focused on indicator objects because that was sufficient to demonstrate the core pipeline.

Conclusion

The project successfully demonstrated an operational threat intelligence workflow connecting OpenCTI with Wazuh. Two external intelligence sources were integrated into OpenCTI, a filtered collection was exposed through TAXII 2.1, and a Python-based consumer maintained a local cache containing IPv4, domain and SHA-256 indicators.

The enrichment pipeline was then connected to Wazuh and validated with controlled security events. The final applied use case extended an existing SSH password-guessing detection rather than replacing it.

The baseline Wazuh rule identified repeated failed SSH authentication attempts. When the source IP matched an indicator in the OpenCTI-derived cache, the workflow produced a second-stage CTI-enriched alert containing the indicator, confidence score, labels, STIX identifier and original detection metadata.

The final result demonstrates how threat intelligence can be operationalized as part of SIEM triage. The project therefore achieved its central goal: showing that CTI can meaningfully add context to a real detection workflow.

References

OpenCTI Documentation <https://docs.opencti.io/latest/>

OpenCTI Deployment Overview <https://docs.opencti.io/latest/deployment/overview/>

OpenCTI Connectors <https://docs.opencti.io/latest/deployment/connectors/>

OpenCTI Data Model <https://docs.opencti.io/latest/usage/data-model/>

OpenCTI Feeds / TAXII <https://docs.opencti.io/latest/usage/feeds/>

OASIS STIX 2.1 Specification <https://docs.oasis-open.org/cti/stix/v2.1/stix-v2.1.html>

MITRE **ATT&CK** – **T1110.001** **Password** **Guessing**
<https://attack.mitre.org/techniques/T1110/001>

MITRE ATT&CK – DET0551 <https://attack.mitre.org/detectionstrategies/DET0551/>

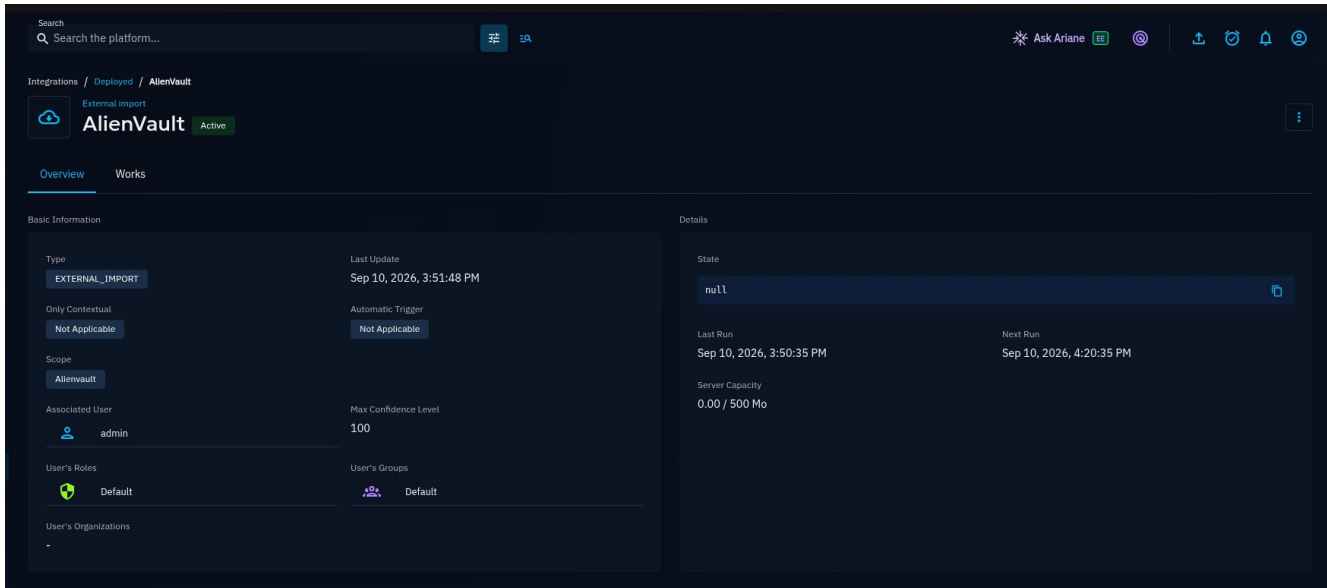
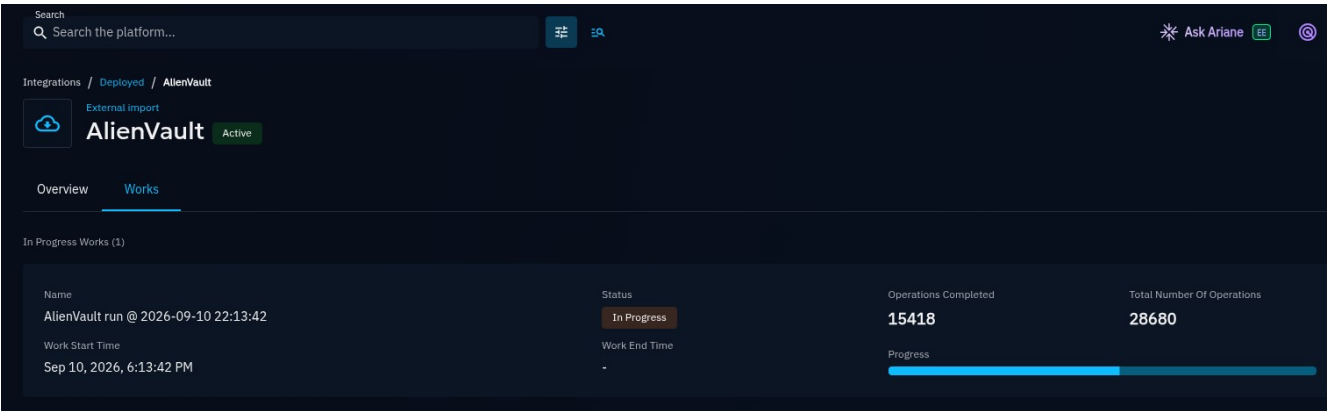
MITRE ATT&CK Navigator <https://mitre-attack.github.io/attack-navigator/>

Wazuh Documentation <https://documentation.wazuh.com/>

AlienVault OTX <https://otx.alienvault.com/>

ThreatFox / Abuse.ch <https://threatfox.abuse.ch/>

Appendix



Search

Search the platform...

Ask Ariane

Techniques / Attack patterns

Search

web-analyzer-serv32.c

Add filter

Select saved filter

1 - 25 / 977

Create Attack Pattern

☐	Kill chain phase	ID	Name	Labels	Original creation date	Modification date	
☐	[mitre-mobile-attack] command-an...	T1437.001	Web Protocols	-	Apr 1, 2022	Sep 8, 2026	>
☐	[mitre-attack-v19] resource-develop...	T1584.006	Web Services	-	Sep 30, 2020	Sep 6, 2026	>
☐	[mitre-mobile-attack] command-an...	T1481	Web Service	-	Feb 1, 2019	Sep 8, 2026	>
☐	[mitre-attack-v19] credential-access	T1606	Forge Web Credentials	-	Dec 16, 2020	Sep 6, 2026	>
☐	-	-	Web Application Fingerprinting	-	Jun 22, 2014	Sep 8, 2026	>
☐	-	-	Upload a Web Shell to a Web Server	-	May 30, 2018	Sep 8, 2026	>
☐	-	-	Detect Unpublished Web Pages	-	Jun 22, 2014	Sep 8, 2026	>
☐	-	-	Web Server Logs Tampering	-	Jun 22, 2014	Sep 8, 2026	>
☐	-	-	Detect Unpublished Web Services	-	Jun 22, 2014	Sep 8, 2026	>
☐	-	-	Poison Web Service Registry	-	Jun 22, 2014	Sep 8, 2026	>
☐	-	-	Web Services Protocol Manipulation	-	Jun 22, 2014	Sep 8, 2026	>
☐	-	-	Cause Web Server Misclassification	-	Jun 22, 2014	Sep 8, 2026	>
☐	-	-	Manipulating Web Input to File System Calls	-	Jun 22, 2014	Sep 8, 2026	>
☐	[mitre-attack-v19] persistence	T1505.003	T1505.003	-	Dec 13, 2019	Sep 10, 2026	>

```
kali@localhost:~/opencti/docker$ sudo docker compose up -d
[+] Running 52/161
  connector-import-document Pulling
  connector-analysis [ ] Pulling
  worker [ ] Pulling
  connector-mitre Pulled
  connector-export-file-csv [ ] Pulling
  elasticsearch [ ] Pulling
  redis [ ] Pulling
  xtm-one [#####] 1.008GB / 1.111GB Pulling
  connector-import-file-stix [ ] Pulling
  opencti [#####] Pulling
  connector-export-file-txt [ ] Pulling
  minio [ ] Pulling
  connector-opencti [ ] Pulling
  connector-export-file-stix [ ] Pulling
  pgsql-xtm-one Pulled
  connector-import-file-yara [ ] Pulling
  rabbitmq [ ] Pulling
  connector-import-external-reference [ ] Pulling
  rsa-key-generator [ ] Pulling
  xtm-composer [ ] Pulling
  xtm-one-worker [ ] Pulling
```

Docker & your first connectors



Search

Search the platform...

Ask Ariane

Threats / Threat actors (group) / Ghost Te7mina1

Ghost Te7mina1

+ add alias

Overview Knowledge Content Analyses Data History

AI Insights

Details

Basic Information

Description

Threat actor created for OpenCTI validation. Used only to verify manual entity creation and relationship functionality.

Threat Actor Types

Activist Hacker Spy

Located At

Ghana

First Seen

-

Last Seen

-

Sophistication

-

Resource Level

-

Primary Motivation

-

Roles

-

Score

16 / 100

Goals

-

Secondary Motivations

-

Marking

-

Author

FBI

Reliability (Of Author)

-

Confidence Level

1 - Confirmed By Other Sources

Distribution Of Opinions

strongly-agree agree disagree neutral

Processing Status

Disabled

Revoked

No

Labels

+

Platform Creation Date

September 9, 2026 at 8:06:45 PM

Creators

Admin

Standard STIX ID

threat-actor--22dc88b0-5a2d-5b62-b6ce-f668da513961

Original Creation Date

September 9, 2026 at 8:06:45 PM

Modification Date

September 9, 2026 at 8:06:45 PM

Latest Created Relationships

Latest Containers About This Object

Search

Search the platform...

Threats / Threat actors (group)

Search

Search these results...

Add filter

Sort by

Name

Create a threat actor group

Create multiple entities

Name*

Ghost Te7/mina1

No potential duplicate entities has been found.

Threat actor types

Hacker X Activist X Spy X

Confidence level

80

1 - Confirmed by other sources

Description

Write Preview H B I

**Threat actor created for OpenCTI validation.
Used only to verify manual entity creation and relationship functionality.**

Paste, drop, or click to add images

Score

16

Author

FBI

+

Labels

+

Markings

External references

+

Associated file

Search

Search the platform...

Observations / Indicators / 104.249.10.86

104.249.10.86

Overview Knowledge Content Analyses Sightings Data History

Details

Basic Information

Description

Traffic seen on port 56003

Indicator Pattern

[ipv4-addr:value = '104.249.10.86']

Valid From

September 10, 2026 at 9:31:01 PM

Valid Until

October 1, 2026 at 3:05:40 AM

Score

50 / 100

Lifecycle

Detection

No

Kill Chain Phases

Indicator Types

IPv4

Main Observable Type

IPv4-Addr

Platforms

Based On

IPv4 Address 104.249.10.86 Sep 10, 2026

Marking

TLP:CLEAR

Author

Threat Fox | Abuse.Ch

Reliability (Of Author)

Confidence Level

2 - Probably True

Distribution Of Opinions

Platform Type

Stix

Processing Status

Disabled

Revoked

No

Labels

PureHVNC X PureRAT X ResolverRAT X botnet_cc X

Platform Creation Date

September 10, 2026 at 9:48:33 PM

Creators

Admin

Standard STIX ID

indicator--000098bf-b647-5e42-95e9-c756c99b5350

Original Creation Date

September 10, 2026 at 9:31:01 PM

Modification Date

September 10, 2026 at 9:48:33 PM

Search

Search the platform...

Ask Ariane

Observations / Indicators / 138a4c9cd617912c2269fae64b6b12d57e926a36c2c62f25ee05a32cdf102212

138a4c9cd617912c2269fae64b6b12d57e926a36c2c62f25ee05a32cdf102212

Create Relationship

Update

Overview

Knowledge

Content

Analyses

Sightings

Data

History

Details

Description

-

Indicator Pattern

{file:hashes, 'SHA-256' = '138a4c9cd617912c2269fae64b6b12d57e926a36c2c62f25ee05a32cdf102212'}

Valid From

September 17, 2026 at 5:02:33 PM

Valid Until

September 14, 2027 at 12:56:18 AM

Score

85 / 100

Lifecycle

Detection

No

Kill Chain Phases

-

Indicator Types

-

Main Observable Type

StixFile

Platforms

-

Based On

File

138a4c9cd617912c2269fae64b6b12d57e926a36c2c62f25ee... Sep 21, 2026

Basic Information

Marking

TLP:CLEAR

Author

AlienVault

Reliability (Of Author)

-

Confidence Level

1 - Confirmed By Other Sou...

Distribution Of Opinions

strongly-disagree

strongly-agree

agree

neutral

disagree

Pattern Type

Stix

Processing Status

Disabled

Revoked

No

Labels

crudeexclude

handala hack

heavygram

social engineering

surveillance backdoor

telegram c2

Platform Creation Date

September 21, 2026 at 10:26:55 AM

Creators

Admin

Standard STIX ID

Indicator--0fea91fc-86f1-5b77-97f9-d889be373c3

Original Creation Date

September 21, 2026 at 10:10:45 AM

Modification Date

September 21, 2026 at 10:26:55 AM

Search

Search the platform...

Ask Ariane

Observations / Indicators

Search

Search these results...

Add filter

Select saved filter

1 - 25 / 14.47K

Create Indicator

Pattern type = and Main observable type =

	Pattern type	Name	Author	Creators	Labels	Original creation date	Marking
☐	Stix	104.249.10.86	Threat Fox Abuse.ch	admin	PureHVCN PureRAT +2	Sep 10, 2026, 9:31:01 PM	TLP:CLEAR
☐	Stix	b1c8a0ddfe6867aa10929e798a13cb249a17e5c47c908fc...	Threat Fox Abuse.ch	admin	RAT asyncrat payload	Sep 10, 2026, 9:30:36 PM	TLP:CLEAR
☐	Stix	web-analyzer-serv32.com	Threat Fox Abuse.ch	admin	CobaltStrike +2	Sep 10, 2026, 9:31:05 PM	TLP:CLEAR
☐	Stix	b691fe33f6868054ae5d9e77b6bb6fe75084ba7a63c85d...	Threat Fox Abuse.ch	admin	RAT asyncrat payload	Sep 10, 2026, 9:30:34 PM	TLP:CLEAR
☐	Stix	hamzasafaraz.com	Threat Fox Abuse.ch	admin	clickfix etherhiding +3	Sep 10, 2026, 9:31:05 PM	TLP:CLEAR
☐	Stix	www.nogaedevloppement.com	Threat Fox Abuse.ch	admin	Etherhiding +2	Sep 10, 2026, 9:30:29 PM	TLP:CLEAR
☐	Stix	38.240.48.155	Threat Fox Abuse.ch	admin	PureHVCN PureRAT +2	Sep 10, 2026, 9:31:00 PM	TLP:CLEAR
☐	Stix	9227d3ae6d90b7c353a2d3c7346c07b84f63fee503aadae...	Threat Fox Abuse.ch	admin	RAT asyncrat payload	Sep 10, 2026, 9:30:44 PM	TLP:CLEAR
☐	Stix	c03f881abed55b7e32c524c740b2628b48819baf08b5bf02...	Threat Fox Abuse.ch	admin	RAT asyncrat payload	Sep 10, 2026, 9:30:31 PM	TLP:CLEAR
☐	Stix	9b471c2935fdd01c7e9d57e78f91d213e6d1b5a44ac1719...	Threat Fox Abuse.ch	admin	RAT asyncrat payload	Sep 10, 2026, 9:30:42 PM	TLP:CLEAR

Search

Search the platform...

Ask Ariane

Observations / Indicators

Search

Search these results...

Add filter

Select saved filter

1 - 1 / 1

Create Indicator

Pattern type = and Main observable type =

	Pattern type	Name	Author	Creators	Labels	Original creation date	Marking
☐	Stix	Ghost Te7mina1 Test-IP	FBI	admin	-	Sep 9, 2026, 8:17:41 PM	-

Search

Search the platform...

Threats / Threat actors (group) / Ghost Te7mina1

Ghost Te7mina1

+ add alias

Overview

Knowledge

Content

Analyses

Data

History

Total analyses → 0 (30 days)

0

Total indicators → 0 (30 days)

0

Distribution Of Reports

No data has b

Diamond

Timeline

Global kill chain

Aliases

Last attributions

View all

Advers

Last used attack patterns

Create a relationship

Threat Actor Group

Ghost Te7mina1

→

Indicator

Ghost Te7mina1 Test-IP

Relationship type

related to

Confidence level

100

1 - Confirmed by other sources

Start time

2026-09-09 08:32 PM

Stop time

2026-09-09 08:32 PM

Description

Write Preview H B I S P Q Y A

Paste, drop, or click to add images

Author

Markings

External references

138a43cd617912c220f6e64bb12d57e926a36c2c62f25e05a32cdf102212

14 / 68

Community Score

138a43cd617912c220f6e64bb12d57e926a36c2c62f25e05a32cdf102212

Size 390.78 MB

Last Analysis Date 19 hours ago

14/68 security vendors flagged this file as malicious

peexe

64bits

detect-debug-environment

overlay

file

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY

Crowdsourced YARA rules

Matches rule SUSP_Imphash_Mar23_3 from ruleset gen_imphash_detection at https://github.com/Neo23x0/signature-base by Armin Rupp (https://github.com/ruppde)

Detects imphash often found in malware samples (Maximum 0.25% hits with search for 'imphashx.pro' on VirusTotal) = 99.75% hits - 1 year ago

Matches rule Pyinstaller from ruleset Pyinstaller at https://github.com/bartblaze/yara-rules by @bartblaze

Identifies executable converted using Pyinstaller. This rule by itself does NOT necessarily mean the detected file is malicious. - 1 year ago

Crowdsourced Sigma Rules

Matches rule Potential Python DLL SideLoading by Swachchhanda Shrawan Poudel at Sigma Integrated Rule Set (GitHub)

Detects potential DLL sideloading of Python DLL files.

Popular threat label

Trojan.python/filelessmalware

Threat categories

trojan

Family labels

python filelessmalware misc

Security vendors' analysis

Alibaba

Trojan.Python/Agent.Rc37662

AliCloud

Trojan.Python/Agent.A6.1

Avast

FilelessMalware [Misc]

AVG

FilelessMalware [Misc]

Fortinet

W32/Agent.AFKNtr

K7AntiVirus

Trojan (005dc3461)

RTGW

Trojan (005dc3461)

Kaspersky

Trojan.Python.Agent.tdu

Do you want to automate checks?

web-analyzer-serv32.com

Did you intend to search across the file corpus instead? Click here

14 / 99

Community Score

web-analyzer-serv32.com

Registrar TUCOWS.COM, CO.

Creation Date 2 months ago

Last Analysis Date 2 days ago

14/99 security vendors flagged this domain as malicious

Spam, Suspicious, Newly Registered (alphaMountain.ai)

spyware and malware

Phishing and Other Frauds

DETECTION

DETAILS

RELATIONS

COMMUNITY

Security vendors' analysis

Do you want to automate checks?

AlphaSOC

Malware

BitDefender

Phishing

Certego

Malicious

Chong Lua Dao

Malicious

CRDF

Malicious

CyRadard

Malicious

G-Data

Phishing

Kaspersky

Malware

Lionic

Malicious

Seclookup

Malicious

SOCRadard

Malware

Sophos

Malware

VIPRE

Malware

Webroot

Malicious

alphaMountain.ai

Spam

ESET

Suspicious

Fortinet

Spam

Gridinsoft

Suspicious

Abusix

Clean

Acronis

Clean

31